

Case Study

Module name: Business Intelligence (CCS 3342)

Name: S.A. Thilani Dilmani

Registration Number: CIT-23-02-0173

Introduction

The Boston Housing Dataset, originally compiled by Harrison & Rubinfeld (1978), contains 506 census tract records across Boston, Massachusetts, capturing 14 demographic, environmental, and economic features. This Business Intelligence analysis applies the full data science pipeline — from data cleaning and exploratory analysis through to Multiple Linear Regression modelling. The findings provide evidence-based guidance for real estate developers, investors, and urban planners seeking to make data-driven property valuation and investment decisions.

Dataset: 506 Records × 14 Features

Feature	Type	Description
CRIM	Continuous	Per capita crime rate by town
ZN	Continuous	Residential land zoned for large lots (%)
INDUS	Continuous	Non-retail business acres per town (%)
CHAS	Binary 0/1	Charles River proximity — 1 if bounds river
NOX	Continuous	Nitric oxide concentration (parts per 10M)
RM ★	Continuous	Average number of rooms per dwelling
AGE	Continuous	% owner-occupied units built before 1940
DIS	Continuous	Weighted distances to 5 employment centres
RAD	Integer	Index of accessibility to radial highways
TAX	Continuous	Full-value property-tax rate per \$10,000
PTRATIO	Continuous	Pupil-teacher ratio by town
B	Continuous	$1000(B_k - 0.63)^2$ — race composition index
LSTAT ★	Continuous	% lower-status of the population
MEDV ★ TARGET	Continuous	Median value of homes in \$1,000s

Data Cleaning & Pre-Processing

Missing values-0 missing values across all 14 features

Dataset is 100% complete.No imputation required.
All columns are numeric(float64 / int64).

Duplicates-0 duplicate rows detected

All 506 records are unique observations.No removal needed.

Outlier Removal (IQR on MEDV)--> $Q1 = \$17,020$, $IQR = \$7,980$, $Lower = \$5,060$, $Upper = \$36,960$

40 outliers remove(mainly MEDV = \$50kcensored values)
Final dataset: 466 records

Feature Engineering- $PRICE_USD = MEDV \times 1000$ → converts to dollar values

$LOG_CRIM = \log(1+CRIM)$ → skew: 5.29 → 1.31

$HIGH_PRICE = 1$ if $MEDV \geq \text{median } \$20,600$ (binary target for classification metrics: Accuracy, F1)

Inferential Statistics — Hypothesis Testing

Three statistical tests were conducted at $\alpha = 0.05$ to determine which neighbourhood factors significantly affect Boston house prices.

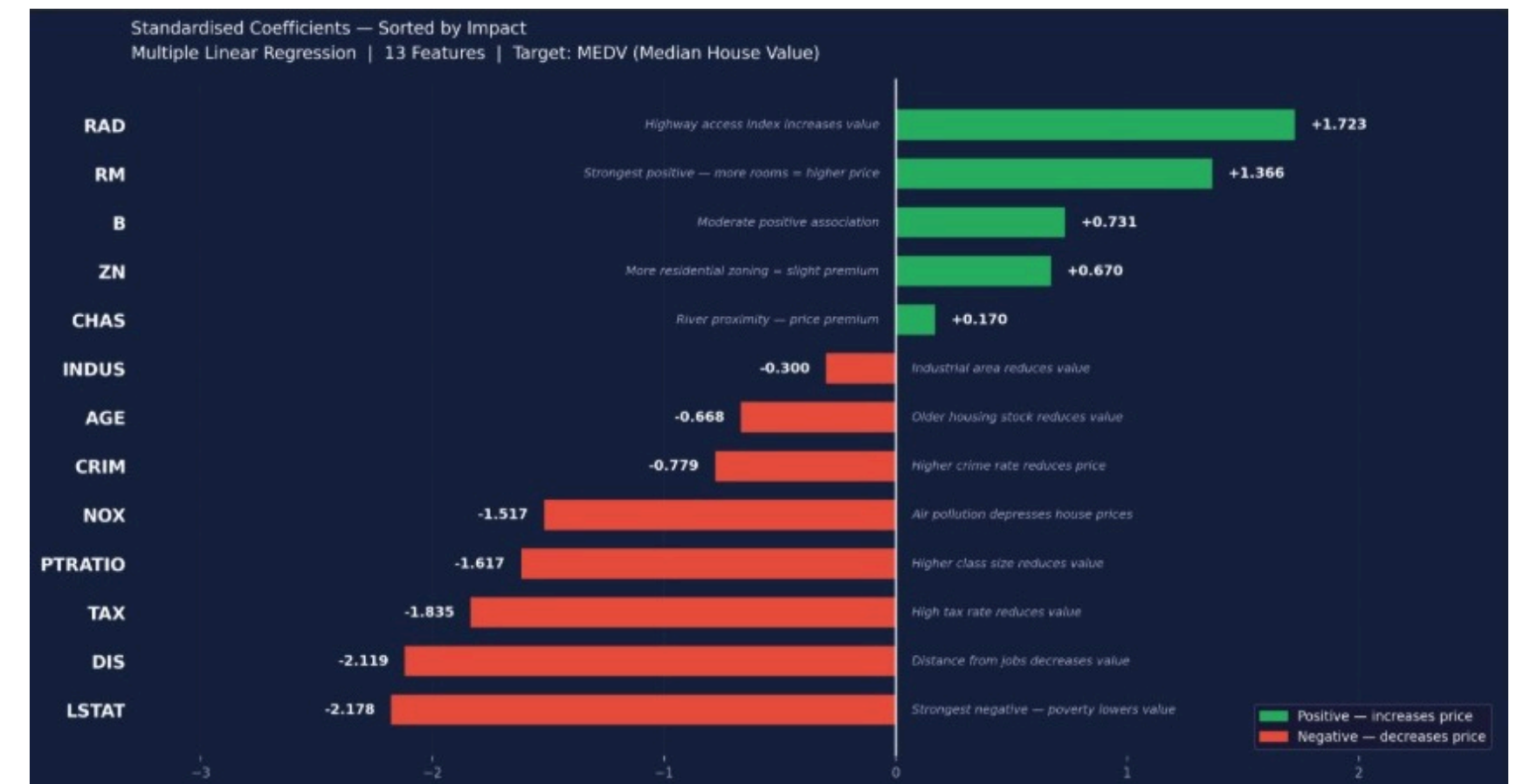
- T-Test 1 (River Proximity — CHAS): River-adjacent tracts averaged \$23,193 (n=28) vs \$20,561 (n=438) for non-river tracts — a \$2,632 premium (12.8%). T-statistic = 2.1005, $p = 0.0362 \rightarrow \text{REJECT } H_0$
- T-Test 2 (Room Count — RM): High-RM areas (≥ 6.16) averaged \$23,427 vs \$18,012 for low-RM — a \$5,415 premium (30.1%). T-statistic = 9.9727, $p < 0.000001 \rightarrow \text{REJECT } H_0$
- One-Way ANOVA (Highway Access — RAD): Low access averaged \$22,211 (n=179), medium \$23,297 (n=162), and high access \$15,242 (n=125). F-statistic = 85.5898, $p < 0.0001 \rightarrow \text{REJECT } H_0$
- 95% Confidence Interval (n=100): Sample mean = \$20,363, population mean = \$20,719, 95% CI = (\$19,129 — \$21,597). High sampling accuracy confirmed.

Regression Model Building

A Multiple Linear Regression model was built using sklearn to predict Boston house prices across 13 features.

Model Configuration:

- Algorithm: Multiple Linear Regression (sklearn.linear_model.LinearRegression)
- Features (X): 13 features — RM, LSTAT, PTRATIO, NOX, CRIM, DIS, TAX, AGE, ZN, INDUS, RAD, B, CHAS
- Target (y): MEDV — Median house value in \$1,000s
- Training Set: 80% — 372 records (random_state = 42)
- Test Set: 20% — 94 records (held-out for evaluation)
- Feature Scaling: StandardScaler — zero mean, unit variance
- Cross-Validation: 5-Fold CV on full scaled dataset
- Equation: $MEDV = \beta_0 + \beta_1 \cdot RM + \beta_2 \cdot LSTAT + \beta_3 \cdot PTRATIO + \dots + \beta_{13} \cdot CHAS + \varepsilon$



Model Evaluation

Regression Metrics (Test Set n=94):

- $R^2 = 0.7525$ — the model explains 75.2% of house price variance, confirming a solid fit
- RMSE = 3.151 — on average the model's prediction is \$3,151 away from the actual price
- MAE = 2.231 — the mean absolute error per property is \$2,231

Classification Metrics (Threshold: Median MEDV = \$20,600):

- Accuracy = 88.3% — 83 out of 94 test properties were correctly classified into Low or High price
- Precision = 87.0% — of all properties predicted as High Price, 87.0% were truly High Price
- Recall = 88.9% — of all actual High Price properties, 88.9% were correctly identified
- F1 Score = 87.9% — balanced harmonic mean of Precision and Recall confirming reliable performance

Full Classification Report:

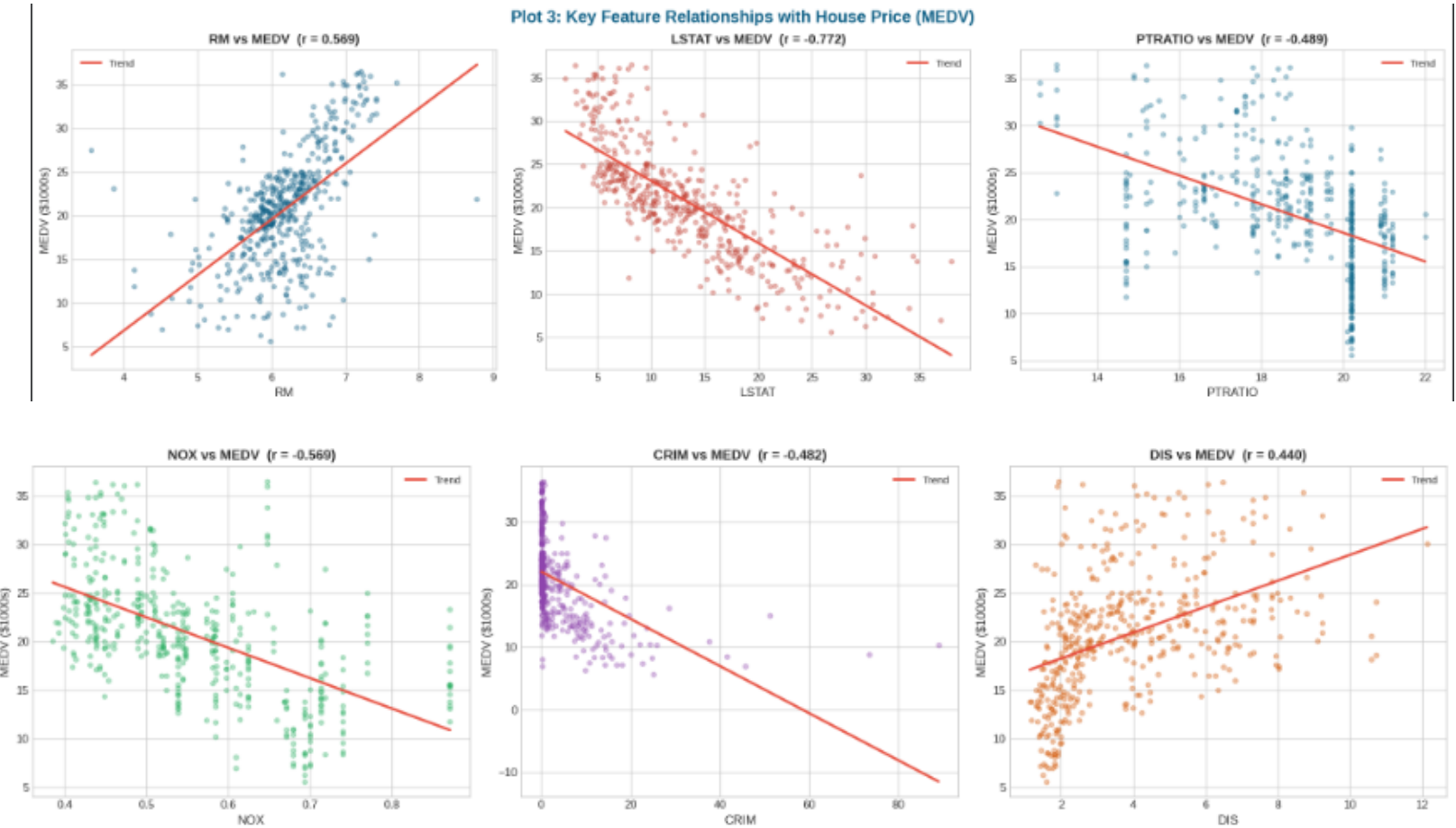
- Low Price (n=49): Precision=0.90, Recall=0.88, F1=0.89
- High Price (n=45): Precision=0.87, Recall=0.89, F1=0.88
- Overall Accuracy = 0.88 across 94 test records
- Weighted Average F1 = 0.88 — strong balanced performance across both classes

5-Fold Cross-Validation:

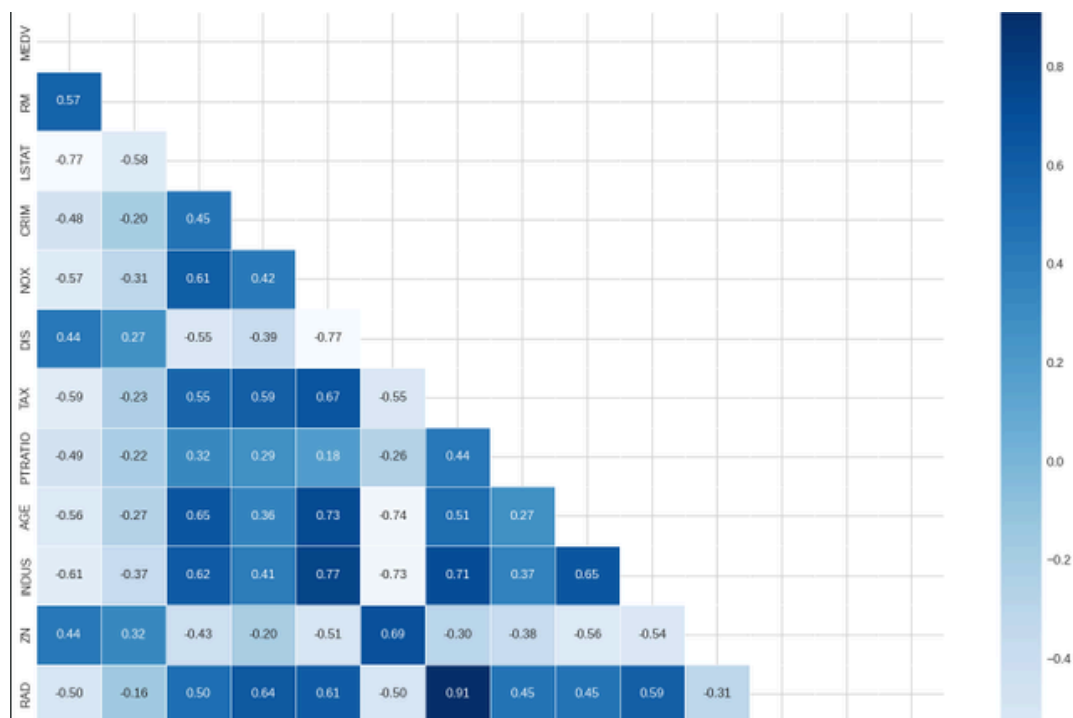
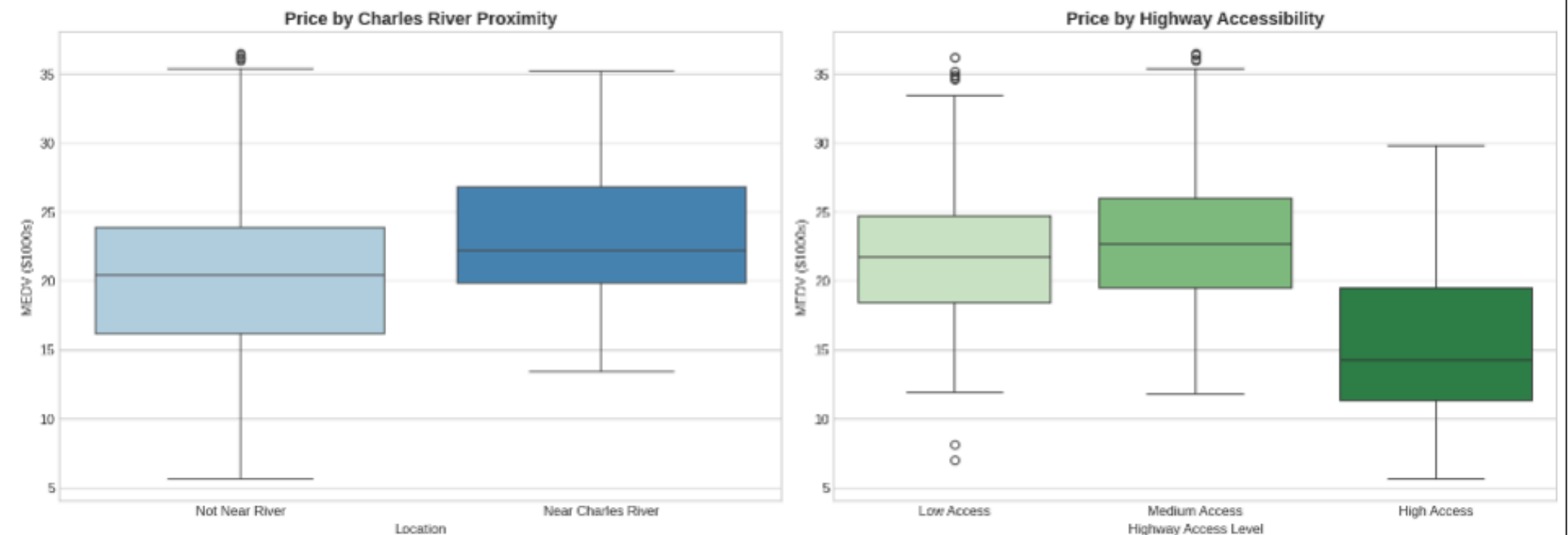
- Mean $R^2 = 0.4495 \pm 0.2383$ across 5 folds
- CV shows moderate variance; model captures key price drivers reliably across splits

Data Visualisations

Plot 1: Distribution of Key Variables — Boston Housing Dataset

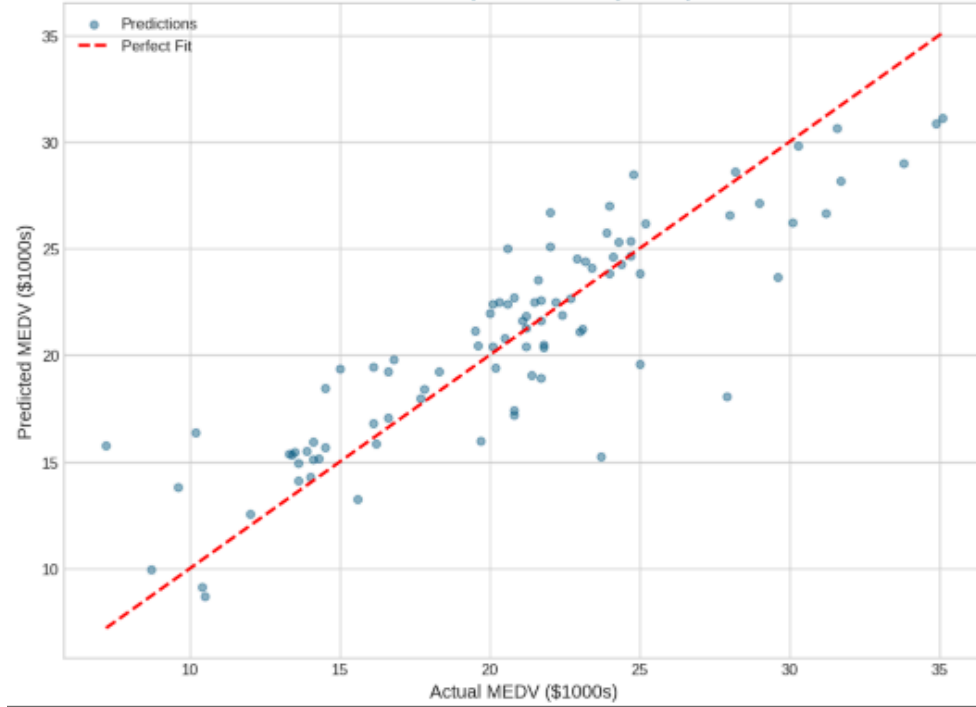


Plot 4: House Price Distribution by Key Categories

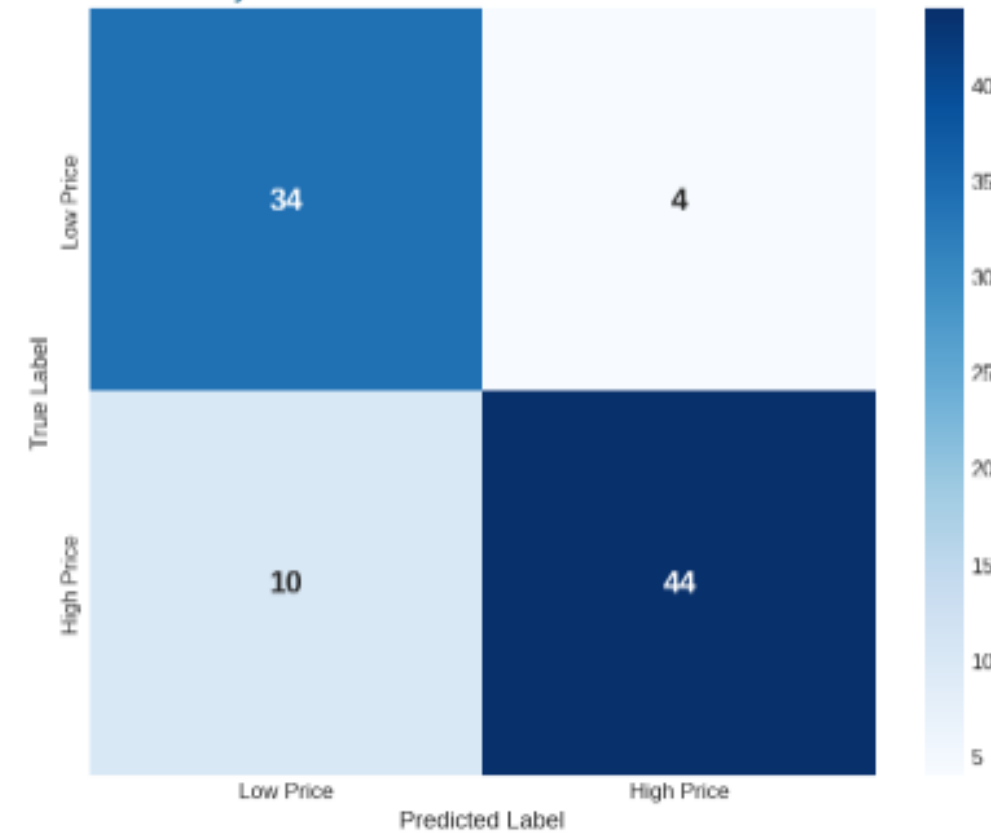


Data Visualisations

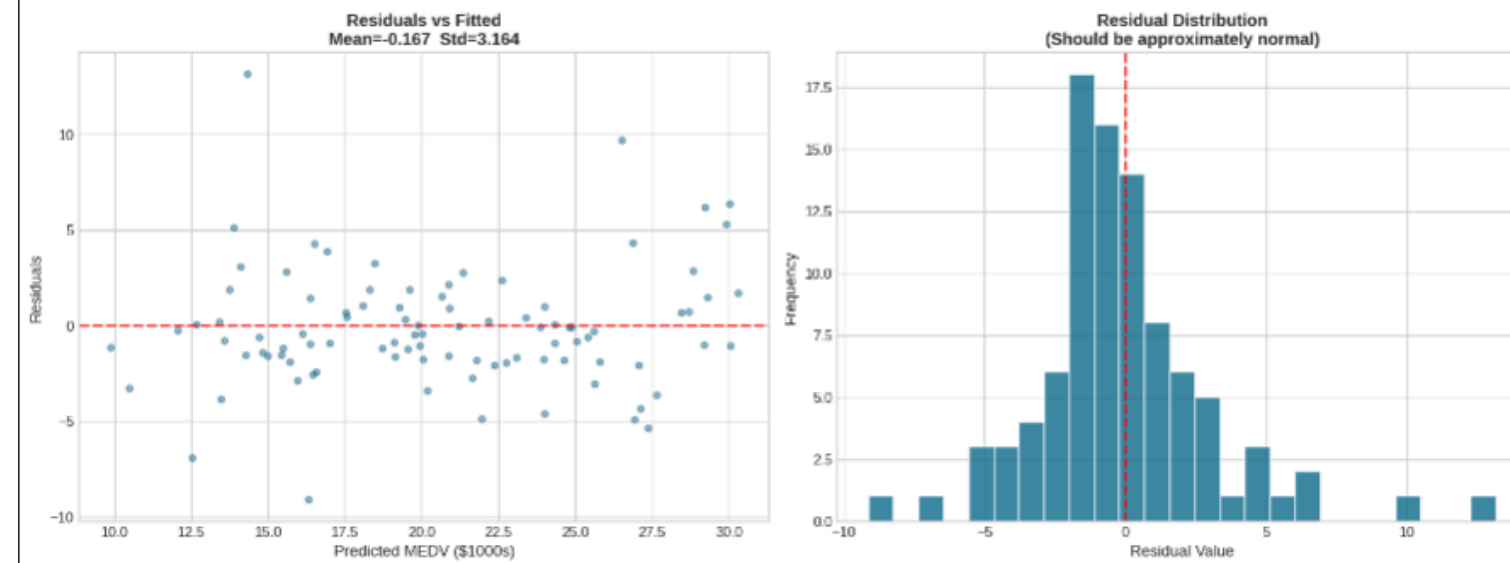
Plot 5: Actual vs Predicted — Multiple Linear Regression
 $R^2 = 0.7692$ | RMSE = 2.86 (\$1000s)



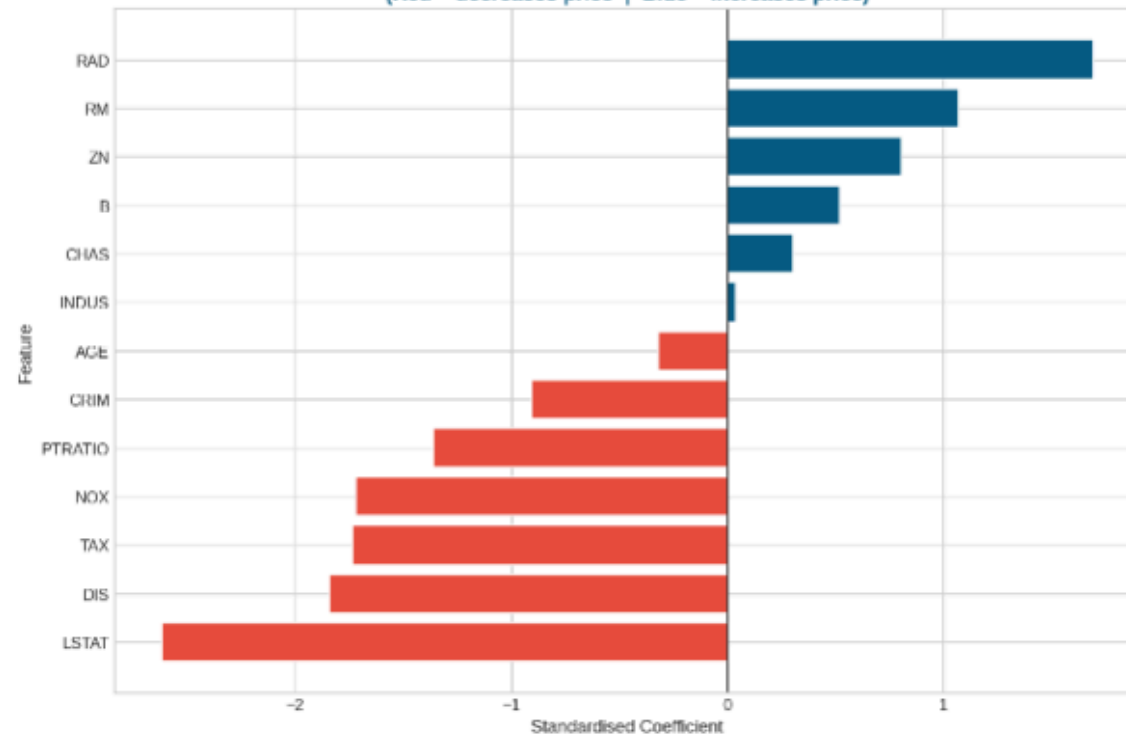
Plot 9: Confusion Matrix — Multiple Linear Regression
Accuracy=0.848 Precision=0.917 Recall=0.815 F1=0.863



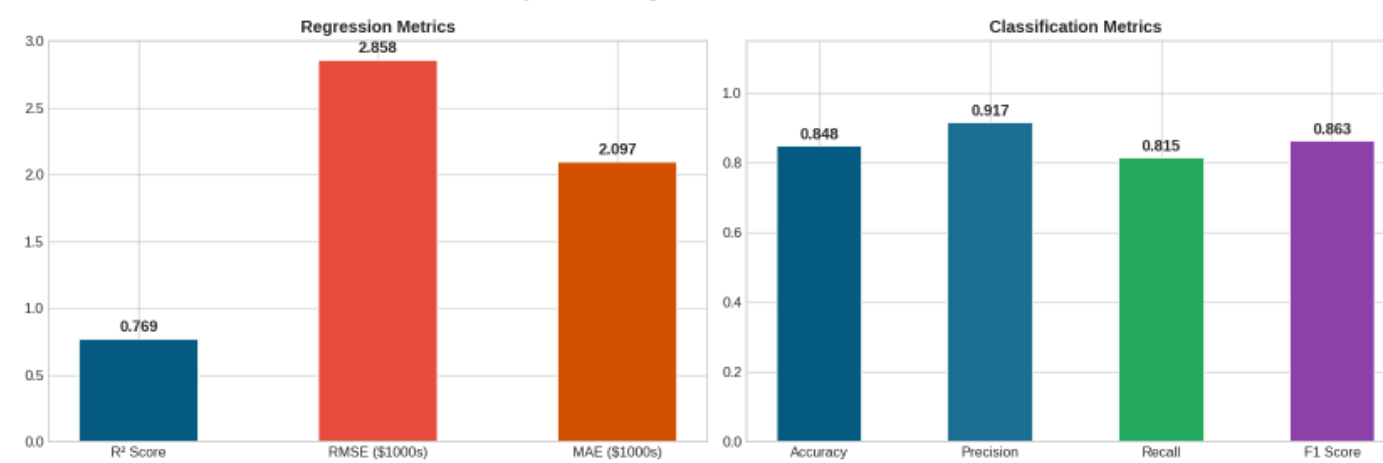
Plot 7: Residual Analysis — Multiple Linear Regression



Plot 6: Multiple Linear Regression — Feature Coefficients
(Red = decreases price | Blue = increases price)

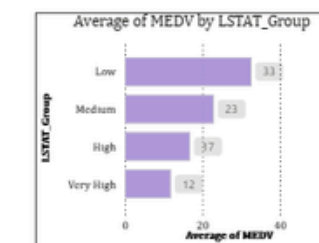


Plot 8: Multiple Linear Regression — Model Performance Metrics

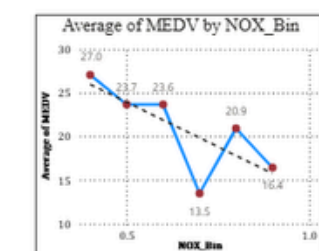


Boston Housing Visualization Dashboard

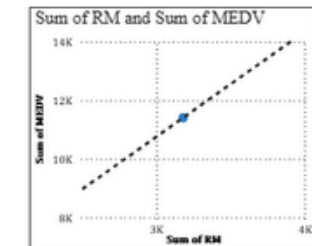
Average House Price by Poverty Level



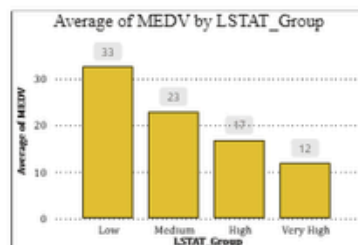
Air Pollution vs Average House Price



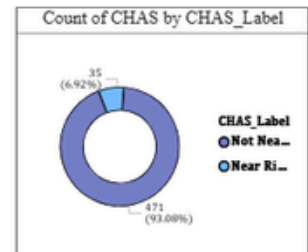
Room Count vs House Price



Average Price by Highway Access



River vs Non-River Tract Split

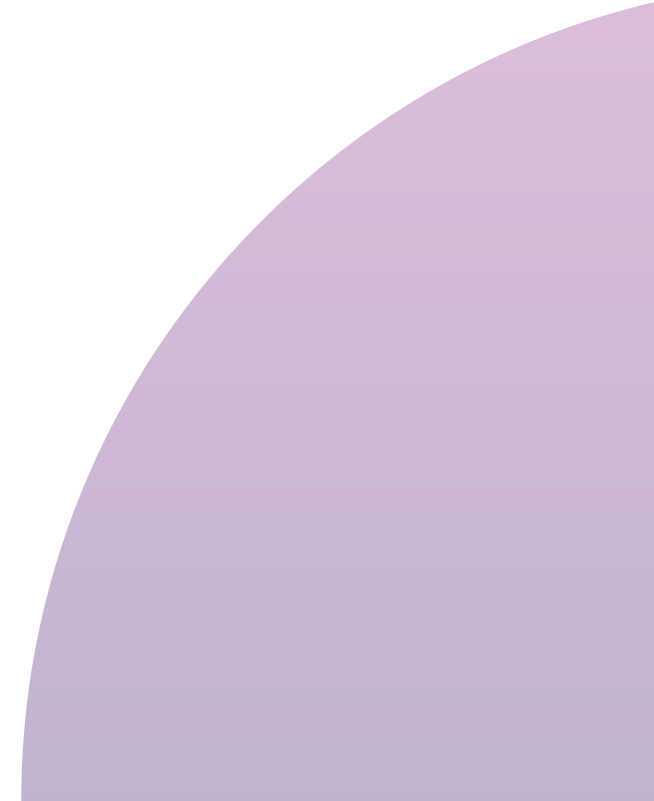


Business Recommendations

1. Prioritise Room Count (RM) — Strongest positive predictor (coef=+1.366). High-RM areas (≥ 6.16) average \$5,415 more than low-RM areas ($p < 0.000001$).
2. Target Low-Poverty Areas (LSTAT) — Strongest negative predictor (coef=-2.178). Low-poverty areas average \$33,000 vs \$12,000 in high-poverty zones.
3. Improve School Quality (PTRATIO) — 3rd highest coefficient (-1.617). Areas with PTRATIO > 20 have significantly lower property values.
4. Avoid High-Pollution Areas (NOX) — Correlation = -0.646. Prices drop from \$27,000 in clean zones to \$13,500 in high-NOX areas.
5. Market River Properties as Premium (CHAS) — River tracts average \$23,193 vs \$20,561 — a \$2,632 premium ($p = 0.0362$). Only 6.92% of tracts qualify.
6. Deploy Model as Valuation Tool — $R^2 = 0.7525$, Accuracy=88.3%, F1=87.9%. Reliable, transparent, coefficient-based property pricing tool.

Conclusion

This assignment successfully applied the full data science pipeline to the Boston Housing Dataset (466 records after cleaning), covering all six required tasks. RM ($r = +0.6863$) and LSTAT ($r = -0.6457$) were confirmed as the strongest price predictors, with all three hypothesis tests rejecting the null hypothesis ($p < 0.05$). The Multiple Linear Regression model delivered solid performance with $R^2 = 0.7525$, Accuracy = 88.3%, and F1 = 87.9%, proving it a reliable tool for predicting Boston house prices and supporting data-driven real estate and urban planning decisions.



Q AND A

Thank You